AI-Powered Cloud Deployment & Recovery Platform

Personal Project Documentation & Technical Overview

Domain: Cloud Computing & DevOps

Submitted By: Himanshu Sonar

Institution: SpeedUp Infotech

Date: September 2026

1. Executive Summary & Abstract

Cloud application deployment is a vital operational process in modern software engineering. However, deployment failures triggered by configuration errors, permission mismatches, service disruptions, and code integration issues often cause extended downtime and require heavy manual intervention.

The **AI-Powered Cloud Deployment & Recovery Platform** automates application deployment on AWS cloud infrastructure while providing intelligent automated recovery and AI-assisted troubleshooting. The system deploys applications directly from GitHub repositories to an AWS EC2 instance using specialized Linux shell scripts. A hybrid troubleshooting engine combines rule-based log checking with the **Google Gemini API** to automatically interpret deployment failures, analyze root causes, and recommend targeted fix steps.

2. Key Technologies Integrated

AWS EC2 | Linux (Ubuntu) | Bash Shell Scripting | Git & GitHub | Python 3 | Google Gemini API | Apache Web Server

3. Problem Statement

In traditional cloud infrastructure workflows, application deployment and error recovery require continuous manual supervision. When a deployment fails, DevOps engineers must manually SSH into instances, inspect multi-megabyte log files, diagnose root causes, and execute manual recovery commands.

Common Deployment Failures

- Linux file and folder permission mismatches.
- Apache web server service crashes or port conflicts.
- Git pull and repository clone failures.
- Missing software dependencies or environment variables.

Operational Friction

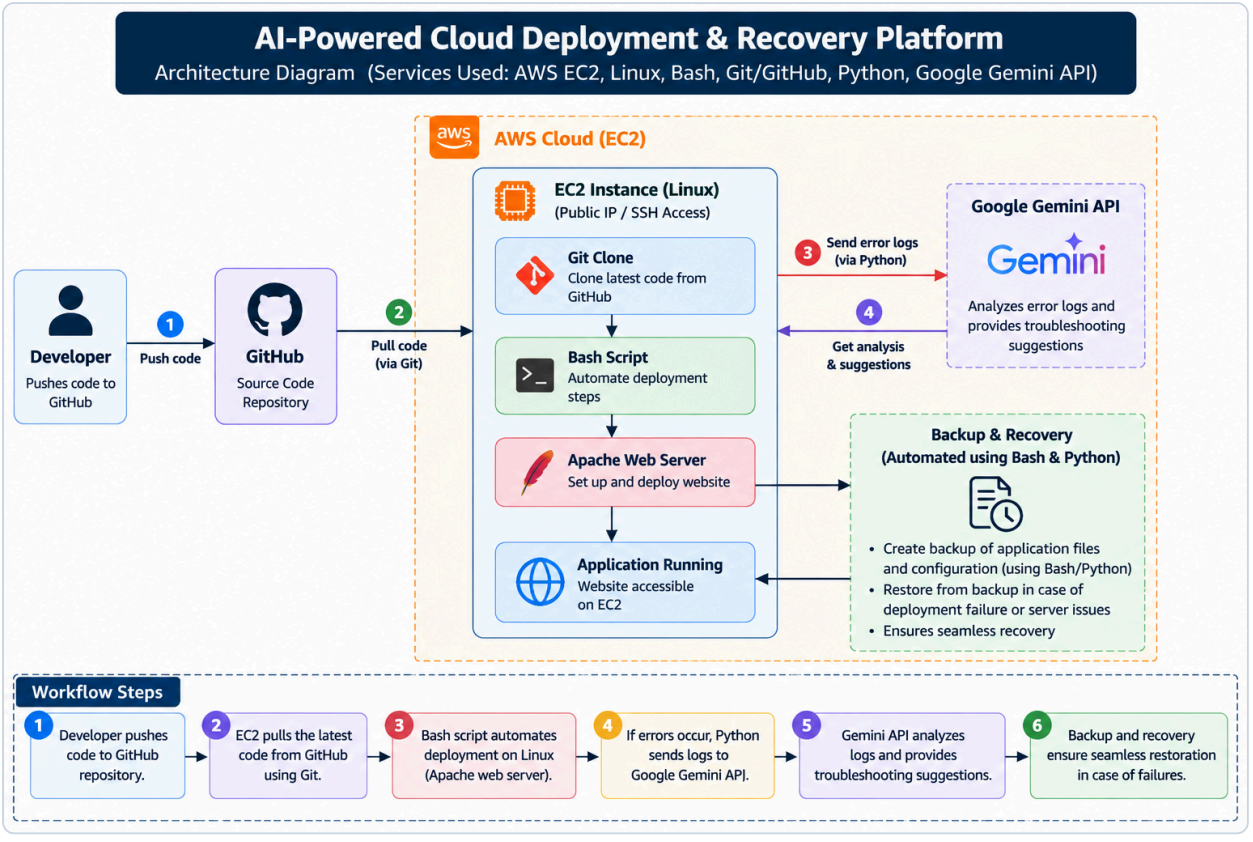
- Extended system downtime during server outages.
- High dependency on skilled DevOps personnel for diagnosis.
- Slow, manual log parsing and troubleshooting processes.
- Absence of fast, automated local recovery mechanisms.

4. Project Objectives

- **Automate Cloud Deployment:** Engineer zero-touch website deployment to AWS EC2 using Linux Bash scripts integrated with GitHub.
- **AI-Driven Log Analysis:** Integrate Google Gemini API to analyze deployment logs automatically and recommend exact corrective actions.
- **Automated Disaster Recovery:** Build automated local backup and system restoration functionality using Bash and Python to ensure high availability and minimal downtime.
- **DevOps Best Practices:** Showcase modular shell scripting, robust error handling, and cloud server administration.

5. System Architecture

The architecture combines automated code pulling from GitHub, deployment orchestration via Bash and Apache on AWS EC2, AI log interpretation via Google Gemini API, and automated backup and recovery mechanisms.



6. Core Functional Modules

1. Automated Deployment Engine

Driven by a master Bash script (`deploy.sh`). It pulls the latest production code from GitHub, sets folder permissions, updates web dependencies, and restarts Apache web services on AWS EC2.

2. AI Log Troubleshooting Engine

Combines local shell error detection with a Python handler connected to the **Google Gemini API**. Parses stack traces, extracts error codes, and outputs actionable fix instructions.

3. Local Backup System

Executed via `backup.sh` prior to every new deployment. Generates timestamped compressed archives of web root directories and configuration files stored locally on EC2.

4. One-Touch Disaster Recovery

Managed by `restore.sh`. If a deployment fails or critical errors are detected, the system immediately restores the last known stable backup, eliminating downtime.

7. Technical Implementation Details

Component	Technology / Tool	Primary Responsibility
Cloud Host	AWS EC2 (Ubuntu Linux)	Virtual server hosting web application & scripts.
Version Control	Git & GitHub	Source code repository and trigger for deployments.
Web Server	Apache HTTP Server	Serves static/dynamic application traffic.
Automation Scripts	Linux Bash Shell (`.sh`)	Handles orchestration, backup creation, and restoration.
AI Integration	Python 3 + Google Gemini API	Performs automated root-cause analysis on logs.

8. Key Project Outcomes & Deliverables

- **Fully Automated Deployment Pipeline:** Reduced manual deployment tasks to a single automated command script execution.
- **Intelligent Failure Recovery:** Instant recommendations and error explanations using Google Gemini API.
- **Robust Disaster Preparedness:** Zero data loss during test failures thanks to automated pre-deployment backups and one-command system rollback.

9. Conclusion

The **AI-Powered Cloud Deployment & Recovery Platform** presents a practical, cloud-native solution combining DevOps automation with artificial intelligence. By streamlining deployment workflows, automating error detection, and ensuring robust backup/recovery on AWS EC2, the project demonstrates a lean, efficient, and industry-relevant architecture for modern web applications.